



Module Name: Neuro.
Module Code: Neu-205
End of Module Exam

Date: 28/11/2024
Time allowed: 120 minutes
Total Marks: 84 marks
Number of pages: 16

The exam consists of (section A) 115 MCQs and (section B) 5 matching to be answered in the bubble sheet

Section A: Identify the choice that best completes the statement or answers the question.

1. Skin over the upper lateral eyelid is supplied by the:

- a- Lacrimal nerve.
- b- Supraorbital nerve.
- c- Supratrochlear nerve.
- d- Zygomaticofacial nerve.

2. Stab wound directed into submandibular triangle may lead to:

- a- Ptosis of the eye.
- b- Drop of the shoulder.
- c- Deviation of the mouth.
- d- Deviation of the tongue.

3. The facial nerve:

- a- Marginal mandibular branch supplies muscles of the upper and lower lips.
- b- Emerges through the stylomastoid foramen.
- c- Has four main branches that exit the parotid gland.
- d- Supplies the anterior belly of digastric.

4. The Abducent nerve supplies the _____ muscle:

- a- Lateral rectus.
- b- Superior tarsal.
- c- Superior rectus.
- d- Inferior oblique

5. Which muscle helps to open the jaw?

- a- Medial pterygoid.
- b- Lateral pterygoid.
- c- Masseter.
- d- Temporalis.

6. Inferior root of ansa cervicalis is derived from:

- a- C1.
- b- C2 and C3.**
- c- C4 and C5.
- d- C6 and C7.

7. The posterior triangle is divided into two triangles by ____ muscle:

- a- Digastric.
- b- Trapezius.
- c- Omohyoid.**
- d- Mylohyoid.

8. A 54-year-old man is admitted to the hospital with severe pain in his nasal cavity. Radiographic examination reveals a carcinoma in his nasal cavity. In which of the following locations would the carcinoma block the hiatus of the maxillary sinus?

- a- Inferior meatus.
- b- Middle meatus.**
- c- Superior meatus.
- d- Nasopharynx.

9. A 37-year-old patient has an infectious inflammation of the dural venous sinus closest to the pituitary gland and a secondary thrombus formation. Which of the following is the most likely site of infection?

- a- Straight sinus.
- b- Cavernous sinus.**
- c- Superior petrosal sinus.
- d- Sigmoid sinus.

10. Which one of the following muscles forms the diaphragm of the oral cavity?

- a- Digastric.
- b- Mylohyoid.**
- c- Geniohyoid.
- d- Stylohyoid.

11. The common site of nose bleeding (Little's area) is located at which part of the nose.

- a- Roof.
- b- Floor.
- c- Septum.**
- d- Left lateral wall.

12. Branches of the carotid artery include:

- a- Inferior thyroid artery.
- b- Ascending pharyngeal artery.**
- c- Internal thoracic.
- d- Superficial cervical.

13. The middle meningeal artery is a branch of the:

- a- Internal carotid artery.
- b- Middle cerebral artery.
- c- Maxillary artery.**
- d- Deep temporal artery.

14. Which one of the following muscles of the palate is innervated by the trigeminal nerve?

- a- Palatoglossus.
- b- Musculus uvulae.
- c- Palatopharyngeus.
- d- Tensor palati.

15. After having a tonsillectomy, a 57-year-old man with a long history of chewing tobacco use is unable to detect taste on the posterior one-third of his tongue. Which of the following nerves has most likely been injured?

- a- Internal laryngeal nerve.
- b- Lingual nerve.
- c- Glossopharyngeal nerve.
- d- Greater palatine nerve.

16. Following a penetrated injury in the submandibular triangle, the tongue of a 45-year-old patient deviates to the left on protrusion. Which of the following nerves is injured?

- a- Right lingual nerve.
- b- Left lingual nerve.
- c- Right hypoglossal nerve.
- d- Left hypoglossal nerve.

17. The ~~superficial~~ ^{Deep} cervical lymph nodes are located alongside the _____ vein:

- a- Subclavian.
- b- Internal jugular.
- c- External jugular.
- d- Anterior jugular.

18. The pharyngeal muscle that originates from the pterygomandibular raphe is the:

- a- Stylo-pharyngeus.
- b- Middle constrictor.
- c- Palato-pharyngeus.
- d- Superior constrictor.

19. The tympanic membrane is at the _____ wall of the middle ear:

- a- Medial.
- b- Lateral.
- c- Inferior.
- d- Anterior.

20. The superior thyroid artery is closely related to the _____ nerve:

- a- Lingual.
- b- Hypoglossal.
- c- ~~Internal~~ laryngeal. ^{External}
- d- Glossopharyngeal.

21. The internal laryngeal nerve supplies:

- a- Sensation to the vocal folds.
- b- Cricothyroid muscle.
- c- Sensation above the vocal folds.
- d- Transverse arytenoid muscle.

22. The _____ muscle of the tongue is innervated by the pharyngeal plexus:

- a- Hyoglossus.
- b- Styloglossus.
- c- Palatoglossus.
- d- Intrinsic vertical.

23. The internal jugular vein:

- a- Drains into the subclavian vein on the left.
- b- Lies in the carotid sheath with the carotid artery and the sympathetic trunk.
- c- Lies lateral to the internal carotid artery at the base of the skull.
- d- Lies lateral to the carotid artery in the lower neck.

24. Which triangle of the neck lies between the anterior bellies of digastric muscles?

- a- Occipital.
- b- Posterior.
- c- Muscular.
- d- Submental.

25. Ptosis (dropping of the upper eyelid) is due to paralysis of the smooth muscle part of the _____ muscle:

- a- Superior rectus.
- b- Superior oblique.
- c- Levator palpebrae superioris.
- d- Depressor anguli oris.

26. Which one of the following ligaments is a thickened part of investing layer of the deep cervical fascia?

- a- Thyrohyoid.
- b- Spuraspinous.
- c- Cricothyroid.
- d- Stylomandibular.

27. Moving the head so that the face looks upwards and to the left is a movement produced by the:

- a- Platysma.
- b- Left trapezius only.
- c- Right geniohyoid only.
- d- Right sternocleidomastoid.

28. A patient had lost cutaneous sensation over the angle of the mandible, which nerve is most likely damaged?

- a- Mandibular.
- b- Ophthalmic.
- c- Great auricular.
- d- Auricotemporal.

29. A 23 years old male represented with swelling of parotid region. He was suffering from much pain in parotid region and ear, which increased while chewing. The nerve most likely to be mediating the pain is?

- a- Middle superior alveolar.
- b- Infraorbital.
- c- Glossopharyngeal.
- d- Auriculotemporal.

30. A patient complains of dryness of eyes, there was decreased secretion of lacrimal glands, the most likely ganglion involved is?

- a- Otic ganglion.
- b- Ciliary ganglion.
- c- Pterygopalatine ganglion.
- d- Superior cervical ganglion.

31. What is the cutaneous branch of the cervical plexus that crosses over the sternocleidomastoid muscle?

- a- Transverse cervical.
- b- Great auricular.
- c- Supraclavicular.
- d- Lesser occipital.

→ Horizontally

32. A 2-month-old male infant had a small pit at the anterior border of the sternocleidomastoid muscle, with mucus dripping intermittently from the opening. The pit extended to the tonsillar fossa as a branchial fistula. Which of the following embryologic structure(s) is (are) involved in this anomaly?

- a- Second pharyngeal arch
- b- Second pharyngeal pouch and groove
- c- Third pharyngeal pouch
- d- Second pharyngeal pouch and cervical sinus

33. A 2-day-old infant male has a noticeable gap in his upper lip. The diagnosis is a cleft lip. Failure of fusion of which of the following structures is the most likely cause of this anomaly?

- a- Maxillary prominences/processes and the intermaxillary segment
- b- Medial nasal prominences/processes
- c- Lateral nasal and medial nasal prominences/processes
- d- Lateral prominences/processes

34. A 45-year-old man is presented to the clinic complaining of loss of general sensation on one side. Which of the following areas is most likely affected?

- a- Area 3, 1, 2
- b- Area 22
- c- Area 17
- d- Areas 18, 19

35. A 55-year-old man is admitted to the emergency department with incontinence of urine and stool. Which of the following areas is most likely affected?

- a- Cuneus
- b- Paracentral lobule
- c- Premotor area
- d- Parietal association area

36. A 67-year-old man is presented to the clinic complaining of inability to talk. Which of the following areas is most likely affected?

- a- Primary olfactory area
- b- Broca's area**
- c- Premotor area
- d- Prefrontal area

37. A 72-year-old man is presented to the clinic complaining that he cannot discriminate the colors. Which of the following areas is most likely affected?

- a- Area 3, 1, 2
- b- Area 17
- c- Areas 41, 42
- d- Areas 18, 19**

38. A 54-year-old woman working as a tailor presents to the clinic complaining that she has not been able to do her work for a few weeks. Which of the following areas is most likely affected?

- a- Areas 40, 39
- b- Area 28
- c- Motor area 4**
- d- Areas 41, 42

39. Which ventricle lies between the pons and cerebellum?

- a- Lateral ventricle.
- b- Third ventricle.
- c- Fourth ventricle.**
- d- Fifth ventricle.

40. Pia matter:

- a- Tough membrane
- b- Is carried into all its sulci**
- c- Bridges over the various sulci on the surface of the brain
- d- Its two layers enclose venous sinuses

41. Cisterna magna is applied to:

- a- Cerebello-medullary cistern**
- b- Cistern of lateral fissure
- c- Interpeduncular cistern
- d- Pontine cistern

42. The part of the diencephalon attached to the posterior part of the thalamus:

- a- Hypothalamus
- b- Epithalamus
- c- Metathalamus**
- d- Subthalamus

43. The part of the internal capsule that contains optic radiation is:

- a- Anterior limb
- b- Posterior limb
- c- Retro-lenticular part**
- d- Sub-lenticular part

44. Fibers connecting corresponding areas in the 2 cerebral hemispheres:

- a- Projection fibers
- b- Corpus callosum**
- c- Cingulum
- d- Superior longitudinal fibers

45. The lateral ventricles are located within which part of the brain?

- a- Cerebellum.
- b- Brainstem.
- c- Cerebral hemispheres.**
- d- Diencephalon.

46. One of the following cranial nerves is attached to the posterior aspect of the brain stem:

- a- Abducent nerve
- b- Vagus nerve
- c- Oculomotor nerve
- d- Trochlear nerve**

47. The following statements concern the third ventricle, one of them is correct:

- a- It is the cavity of the hindbrain.
- b- It communicates with the lateral ventricles through the cerebral aqueduct.
- c- It is continuous with the fourth ventricle through the interventricular foramen.
- d- The choroid plexus is located in the roof.**

48. The middle cerebral artery supplies one of the following parts:

- a) Auditory area.**
- b) The sublentiform part of internal capsule.
- c) Motor and sensory areas of the whole body.
- d) Medial half of orbital surface of cerebral hemisphere.

49. The posterior surface of midbrain is represented by:

- a- Upper part of 4th ventricle
- b- Lower part of 4th ventricle
- c- Cerebral peduncle
- d- Corpora quadrigemina**

50. The fissure that separates the flocculonodular lobe from posterior lobe of the cerebellum:

- a- Primary fissure
- b- Horizontal fissure
- c- Posterolateral fissure**
- d- Lateral fissure

51. Concerning the superficial middle cerebral vein choose one best answer:

- a- It lies in the posterior ramus of the lateral sulcus.**
- b- It receives the majority of inferior cerebral veins of the medial surface.
- c- Ends in the transverse sinus.
- d- Ends in the superior sagittal sinus.

52. The arterial supply of the lumbar enlargement of the spinal cord:

- a- The arteria radicularis magna
- b- The anterior spinal artery
- c- The posterior spinal arteries
- d- The basilar artery

53. Presynaptic inhibition is characterized by:

- a- Decreased Cl^- influx into presynaptic terminals
- b- Hyperpolarization of presynaptic terminals
- c- Increased Ca^{2+} influx into post-synaptic neuron
- d- A 3rd neuron secreting serotonin

54. Concerning synaptic transmission:

- a- Depends on presence of voltage-gated Ca^{2+} channels in the postsynaptic membrane
- b- Presynaptic knobs release only an excitatory transmitter
- c- It is physiologically terminated by block of postsynaptic receptors
- d- It is inhibited by hypoglycemia and acidosis

55. The release of neurotransmitter at a chemical synapse in the CNS is dependent on:

- a- Synthesis of acetylcholinesterase
- b- Hyperpolarization of the synaptic terminal
- c- Opening of ligand-gated calcium channel
- d- Influx of calcium into the presynaptic terminal

56. Inhibitory postsynaptic potential (IPSPs):

- a- Interacts with other potentials to move the membrane potential of postsynaptic neuron away or toward firing level
- b- Is produced by decreased Cl^- conductance
- c- Can be produced by an increase in Na^+ conductance
- d- Is produced as a result of the presence of an inhibitory 3rd neuron

57. Post-tetanic potentiation in synapses:

- a- Is due to activation of AMPA and NMDA receptors by glutamate
- b- Is produced by an excitatory 3rd neuron secreting serotonin
- c- Is due to opening of transmitter gated Cl^- channels
- d- Is due to accumulation of calcium in the presynaptic terminal

58. Convergence in a neuronal pool:

- a- Is the state when one neuron stimulates many neurons
- b- Helps in the distribution of signals
- c- Helps in the selection of important signals and ignoring unimportant ones
- d- Involves temporal summation of a weak input fiber along sensory pathways

spatial

59. Facilitation:

- a- Is due to stimulation of 2 input neurons away from each other.
- b- Is due to stimulation of 2 input neurons close to each other.
- c- Number of action potentials released is less than if each input neuron is stimulated alone.
- d- Occurs when the two afferent neurons have a common discharge zone.

60. The receptor potential:

- a- Has a long absolute refractory period
- b- Amplitude is not related to the strength of the stimulus
- c- Always generates an action potential in the receptor
- d- Its duration is longer than action potential duration

61. Regarding sensory receptors:

- a- Stimulus energy is converted into a local depolarization
- b- The receptor potential is graded and self-propagating
- c- The receptor potential can be produced by only one form of energy
- d- The frequency of action potentials generated doubles when the strength of the stimulus doubles

62. Receptors detect increased stimulus intensity by:

- a- Lowering the threshold for receptor stimulation
- b- By generating receptor potentials having higher magnitudes
- c- By generating nerve impulses that are transmitted along sensory fibers at higher velocities
- d- By enhancing the central effects of sensory impulses

63. The ability to localize the site of stimuli depends upon:

- a- The type of the stimulated receptor
- b- The rate of adaptation of the stimulated receptors
- c- Connections between the receptor and the sensory cortex
- d- The nature of the stimulus

64. Thermal receptors are:

- a- Rapidly adapting receptors
- b- Moderately adapting receptors
- c- Are stimulated at zero 0°C
- d- All are encapsulated receptors

65. Pain receptors are:

- a- Similar in structure to paccinian corpuscles
- b- Sensitized by a rise in the local tissue K^{+} concentration
- c- Quick to adapt to a constant stimulus
- d- Stimulated in the wall of the GIT by sharp cutting

66. Visceral pain:

- a- Is poorly localized compared with pain arising in skin
- b- May cause reflex contraction of overlying skeletal muscle
- c- May cause reflex vomiting
- d- All of the above

67. Inhibition of pain signals by tactile stimulation of a skin surface involves:

- a- Type A alpha fibers in peripheral nerves
- b- Type A beta fibers in peripheral nerves
- c- Type A delta fibers in peripheral nerves
- d- Type C fibers in peripheral nerves

68. After anterolateral cordotomy, relief of pain is due to interruption of:

- a- Dorsal column
- b- Ventral spinothalamic tract
- c- Ventral spinocerebellar tract
- d- Lateral spinothalamic tract

69. The neurons that release serotonin as chemical transmitter are located in which part of supraspinal analgesia system:

- a- Periaqueductal gray area
- b- Interneurons of the spinal cord
- c- Periventricular area
- d- Nucleus raphe magnus

70. A lesion of the dorsal column pathway is most likely to affect:

- a- Fine touch
- b- Hearing
- c- Pain sensation
- d- Temperature sensation

71. A subject suffers from right hemisection of the spinal cord shows:

- a- Upper motor neuron paralysis at the level of the hemisection
- b- Loss of pain and temperature on the right side below the level of the section.
- c- Loss of fine touch and vibration sense on the left side below the level of the section.
- d- Loss of all sensations on the same side at the level of the section

72. Tabes dorsalis is:

- a- Due to virus infection of the posterior root ganglia
- b- Accompanied with shuffling gait
- c- Accompanied with incoordination of voluntary movements.
- d- Accompanied with loss of crude touch.

73. Regarding muscle spindles:

- a- Are composed of extrafusal fibers
- b- Gamma efferents supply their peripheral parts
- c- Nuclear chain fibers show dynamic response
- d- Afferent discharge decreased with muscle stretch

74. The central receptor area of nuclear-bag fibers is innervated by:-

- a- A gamma nerve fibers
- b- A beta nerve fibers
- c- An alpha nerve fibers
- d- Ia nerve fibers

75. When a muscle is suddenly stretched; a signal is transmitted over Ia sensory fibers from muscle spindles. Which of the following statements best describes the response elicited by these spindle afferent signals?

- a- Contraction of the muscle in which the active spindles are located
- b- Relaxation of the muscle in which the active spindles are located
- c- Relaxation of intrafusal fibers in the active spindles
- d- Direct synaptic activation of gamma motor neurons

76. Which of the following best describes the Golgi tendon organs?

- a- They are located in parallel to the extrafusal muscle fibers
- b- They are inhibited when the muscle actively contract
- c- They are the receptors for the inverse stretch reflex
- d- They are connected to group Ia afferent fibers.

77. Which of the following describes the urinary bladder of a patient during the stage of spinal shock?

- a- Retention with overflow
- b- Automatic bladder
- c- Hyperactive bladder with mass reflex
- d- Evacuated only by voluntary control

78. Lesion at which level of the spinal cord causes immediate death?

- a- Transection at the upper cervical level
- b- Transection at lower cervical level
- c- Transection in the upper thoracic region
- d- Transection in the lower lumbar region

79. Which statement concerning the hair cells of the crista ampularis is correct?

- a- Movement that bends the stereocilia towards the kinocilium depolarizes the hair cell
- b- Bending of the stereocilia to the kinocilium activates voltage-gated sodium channels
- c- Depolarization of the hair cell is achieved by linear acceleration
- d- They are stimulated with constant speed rotation

80. Which of the following structures is maximally sensitive to linear head movement in the vertical plane?

- a- Macula of the utricle
- b- Macula of the saccule
- c- Crista ampullaris of the anterior semicircular duct
- d- Crista ampullaris of the horizontal semicircular duct

81. Which statement concerning the premotor cortex is correct?

- a- It is located just posterior to the primary motor cortex.
- b- It is facilitatory to muscle tone
- c- It controls pattern of complex coordinated gross movements
- d- It contains highly excitable Betz cells

82. A person who has had a traumatic brain injury seems to be able to understand the written and spoken word but his speech is poorly articulated, produced slowly and with great effort. This person most likely has damage to which area of the brain?

- a- Wernicke's area
- b- Broca's area
- c- Angular gyrus
- d- Auditory association area

83. Hypertonia and hyperreflexia are shown in which of the following conditions?

- a- Hypothyroidism
- b- Sleep
- c- Lower motor neuron lesion
- d- Upper motor neuron lesion

84. In lower motor neuron disease, the subject may show:

- a- Rigidity of muscles.
- b- Hyperactive reflexes
- c- Wasting of muscles.
- d- Clonus.

85. Upper motor neuron lesion is manifested by:

- a- Spastic paralysis.
- b- Muscle wasting.
- c- Normal planter response
- d- Fasciculation

86. Which of the following is a symptom of Parkinsonism?

- a- Kinetic tremors
- b- High steppage gait
- c- Slow monotonous speech
- d- Hypotonia

87. Which of the following statements concerning the general functional role of the cerebellum is correct?

- a- The cerebellum directly stimulates motor neurons required to make a movement
- b- The cerebellum controls muscle activity on the opposite side of the body
- c- The cerebellum plays an active role in the coordination of the muscles required to make a movement
- d- The cerebellum is not involved in the planning of a movement, only its execution

88. Which of the following statements best describes a functional role for the intermediate zone of the cerebellum?

- a- Facilitates muscle tone
- b- Servo comparator function and sends corrective impulses to the cortex
- c- Planning function through connection with the cerebral cortex
- d- Functions with the vestibular nucleus to control equilibrium

89. Neurological disease associated with the cerebellum produces which of the following types of symptoms?

- a- Resting tremor
- b- Akinesia
- c- Rigidity
- d- Ataxia

90. Which of the following statements concerning electroencephalogram activity is correct?

- a- Beta waves occur in normal adults who are awake but in a quiet, resting state
- b- Alpha waves occur during periods of increased brain activity and thinking
- c- Theta waves are recorded during REM sleep
- d- Delta waves are characteristic of slow wave sleep

91. Which of the following is a characteristic finding in REM sleep phase?

- a- Increase muscle tone
- b- Increase heart rate and respiratory rate
- c- The eyes are deviated up
- d- Dreams which are not remembered

92. Which of the following is a major cause of blindness due to increased pressure on the optic nerve?

- a- Cataract
- b- Astigmatism
- c- Glaucoma
- d- Presbyopia

93. All of the following are causes of corneal transparency Except:

- a- Non keratinized surface epithelium
- b- It is avascular
- c- Relative corneal dehydration
- d- It is supplied by myelinated sensory nerve fibers

94. The fovea centralis has the highest visual acuity because:

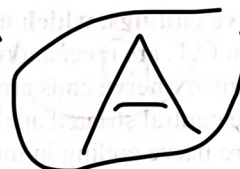
- a- It has condensed rod photoreceptors
- b- Rods are covered by blood vessels
- c- Absence of convergence on the bipolar cells
- d- The pigmented epithelium of fovea is not well developed

95. In dark adaptation:

- a- Retinal sensitivity to light increases
- b- Decreased signal intensity in retinal neurons
- c- Constriction of pupils
- d- Retinal threshold increase

96. Concerning hypermetropia:

- a- The eye is often shorter than the emmetropic eye
- b- The near point distance decreases less than 10 cm
- c- Accommodation for both near and far vision is not needed
- d- Parallel light rays converge in front of the retina



97. Which of the following events makes the auditory system interpret a sound as loud?

- a. Fewer inner hair cells are stimulated
- b. Hair cells excite nerve endings at a diminished rate
- c. Amplitude of vibration of the basilar membrane decreases
- d. Amplitude of vibration of the basilar membrane increases

98. Regarding hair cells:

- a- Hair cells depolarize when their hairs are bent toward the modiolus
- b- Nerve fibers stimulated by hair cells have their cell bodies in the cochlear nuclei of the brain stem
- c- There are more inner hair cells than outer hair cells in the organ of corti
- d- Transmission of auditory signals is performed mainly by inner hair cells than outer hair cells

99. Amplification of sound by the middle ear:

- a. Occurs because the ligaments connecting the bones are usually flexible
- b. Is decreased by cutting the facial nerve
- c. Is caused by the fact that the tympanic membrane is larger in area than the oval window
- d. Occurs because the ossicles act as a lever system to increase the sound 12th times

100. Vitamin A deficiency causes:

- a- Poor night vision
- b- Poor color vision
- c- Astigmatism
- d- Presbyopia

101. Unmyelinated nerve fibres with Schwann cells are found in:

- a- Peripheral nerve fibers.
- b- Post ganglionic sympathetic fibers.
- c- Grey matter.
- d- White matter.

102. Purkinje cells of the cerebellar cortex is an example of:

- a- Bipolar nerve cell.
- b- Multipolar pyramidal nerve cell.
- c- Unipolar nerve cell.
- d- Multipolar pyriform nerve cell.

103. The answer that best applies to neuroglia is:

- a- Spines decorate cell body & branches of mesoglia.
- b- Protoplasmic astrocytes have few long processes.
- c- Microglia have centrioles, so can divide.
- d- Ependymal cells are found around nerve cells in ganglia.

104. Regarding the nerve endings, which of the following statements is true?

- a- Free nerve endings in C.T. are mechanoreceptors for movement & position.
- b- In tendon spindle, sensory nerve ends around the collagen bundles.
- c- Intrafusal fibers have central striated area with no nuclei.
- d- Ruffini corpuscles are nerve ending in muscular tissue.

105. At view of the vestibular apparatus, which statement is correct?

- a- Cristae ampullaris register vertical acceleration & deceleration of the head.
- b- Utricle & saccule are membranous sacs lined by simple columnar epithelium.
- c- Columnar supporting cells have apical stereocilia.
- d- Cells of cristae ampullaris are covered by (cupula) that lacks CaCO_3 crystals.

106. Reissner's (vestibular) membrane forms the:

- a- Outer wall of the cochlear duct.
- b- Roof of the cochlear duct.
- c- Floor of the cochlear duct.
- d- Spiral lamina.

107. Regarding the organ of Corti, which of the following is correct:

- a- Neuroreceptor for equilibrium
- b- Inner phalangeal cells surround inner hair cells completely.
- c- Outer hair cells are arranged as one row.
- d- Border cells support the outer edge of organ of Corti.

B

108. In retina, which of the following is correct regarding the Rod cells?

- a- Its inner fiber ends by pedicle.
- b- Its discs are continuously shed off and phagocytosed.
- c- Located mainly at center of retina.
- d- Slight constriction between inner segment and cell body.

109. After the corneal transplantation, the following structural features provides the best explanation for why corneal graft is not rejected by recipient:

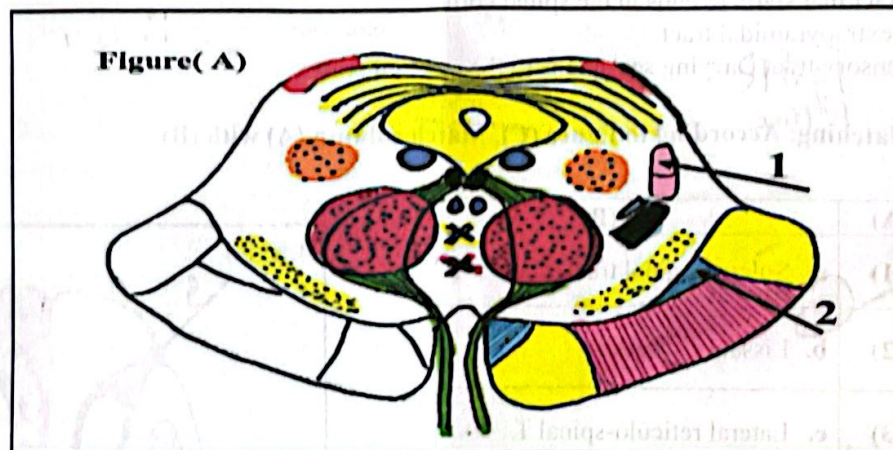
- a- Basal cells of the corneal epithellum are able to regenerate.
- b- The Bowman membrane protects against the spread of infections.
- c- Collagen fibers are regularly packed.
- d- Cornea is avascular and lacks lymphatic drainage.

110. Microscopic examination of corneal epithellum revealed the following:

- a- The superficial layer has microvilli
- b- Keratinized stratified squamous epithellum
- c- Many myelinated nerve fibres
- d- Few mitotic figures

111. Concerning the Iris, the following is correct:

- a- Stroma is formed of loose vascular pigmented C.T.
- b- Fibroblasts and melanocytes cover the posterior surface
- c- Pigmented epithellum covers the anterior surface
- d- It is the posterior part of the vascular coat



According to Figure (A), answer the following questions (112 & 113)

112. Regarding structure pointed out by number (1), choose the correct answer:

- a- It originates from gracile & cuneate nuclei in closed medulla.
- b- Formed by union of lateral & ventral spinothalamic tracts.
- c- Terminates in anterior horn cells (AHCs) of spinal cord.
- d- An associative bundle of ascending & descending myelinated fibers.

113. Select the best statement that applies to structure pointed to by number (2):

- a- It ends on motor nuclei of cranial nerves 3,4 & 6 of opposite side.
- b- At lower part of closed medulla, 80-90% of its fibers cross to opposite side.
- c- It arises from the brain stem to ends in anterior horn cells of the spinal cord.
- d- Originates from lower 1/3 of motor area 4 in cerebral cortex.



According to the figure (B), answer the following questions (114 & 115)

114. As regards the structure pointed out by number (1), find the correct answer:

- a- Its second order neuron is the cells of dorsal root ganglia.
- b- Originates from cells of SGR of opposite side.
- c- Carries crude touch from opposite side.
- d- Responsible for spino-visual reflex.

115. The structure pointed by number (2) is:

- a- A sensory tract that reaches the subcortical level.
- b- A short tract that starts & ends in the spinal cord.
- c- A single extrapyramidal tract.
- d- A long sensory tract carrying sensations to the cerebral cortex.

Section B: Matching: According to figure (C), Match column (A) with (B)

(A)	(B)
1-Label (1)	a. Sulcomarginal tract (T.) 3
2-Label (2)	b. Lissauer's tract 5
3-Label (3)	c. Lateral reticulo-spinal T.
4-Label (4)	d. MSN 2
5-Label (5)	e. SGR
	f. Gracile tract 1
	g. Lateral vestibulo-spinal T. 4

